

Course Code: 20MCA188**Course Name: ARTIFICIAL INTELLIGENCE**

Max. Marks: 60

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 Define an agent and rational agent in AI. (3)
- 2 Solve the following crypt arithmetic problem. (3)

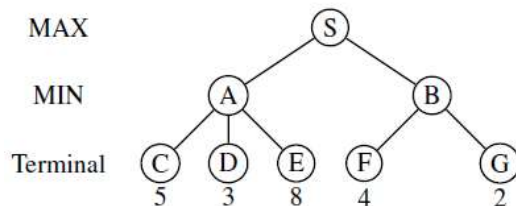
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(3)

- 3 Compare and contrast BFS and DFS methods. (3)
- 4 Define a heuristic function and an admissible heuristic function with examples. (3)
- 5 Compute MINIMAX(S) in the following game tree. (3)



(3)

- 6 List the requirements for knowledge representation systems in AI. (3)
- 7 Explain the inference rules in FOPL. (3)
- 8 Describe supervised, unsupervised and reinforcement learning. (3)
- 9 Give a short note on role of an expert system (3)
- 10 List some membership functions that define a certain special fuzzy sets. (3)

PART B*Answer any one question from each module. Each question carries 6 marks.***Module I**

- 11 Solve missionaries and cannibals problem. (6)

OR

- 12 Describe a production system in AI. What are the merits and demerits of production systems? (6)

Module II

- 13 Explain various uninformed search strategies. (6)

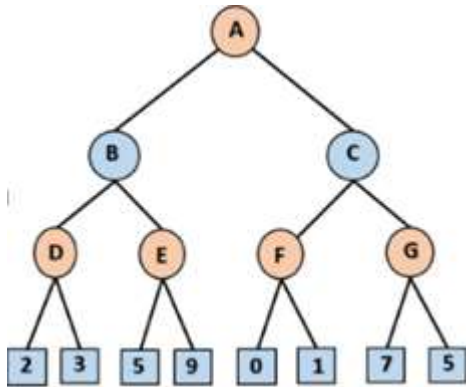
OR

- 14 Explain the following types of hill climbing: (6)

- Simple hill climbing.
- Steepest-ascent hill climbing

Module III

- 15 Explain alpha-beta pruning and determine which of the branches in the game tree below will be pruned if we apply alpha-beta pruning to solve the game (Assume that the maximising player plays first). (6)



OR

- 16 a) Describe the different types of semantic networks with examples. (3)
b) List advantages and disadvantages of semantic networks. (3)

Module IV

- 17 Explain the algorithm to convert WFF to clause with an example (6)

OR

- 18 Explain Neural net and Genetic learning methods in AI (6)

Module V

- 19 Briefly explain about typical expert systems. (6)

OR

- 20 Given the fuzzy sets (6)
 $A = \{0.3/2, 0.4/3, 0.1/4, 0.8/5, 1.0/6\}$
 $B = \{0.7/4, 0.5/5, 1.0/6, 0.02/7, 0.75/8\}$ find $A \cup B$ and $A \cap B$.
